Lecture 30: Diffusion IV

Text to image: conditioning and guidance

Some panels are stills from Welch Labs / 3Blue1Brown (2025).

Everything so far has been unconditional

We can start from noise and land on the data. The digits in L29 were digits - but we could not ask for a **seven**.

Every model in this chapter has drawn *a* sample from $p(\mathbf{x})$. What people actually want is a sample from

$$p(\mathbf{x} \mid \text{"a lone tree in the desert"})$$

Two problems, and they are separate:

1. How does a **sentence** become something a convolutional network can use?
2. Once it is in there, how do we make the model actually **obey** it?

The second turns out to be the harder one, and the answer is surprisingly blunt.

Outline

One space for words and pictures

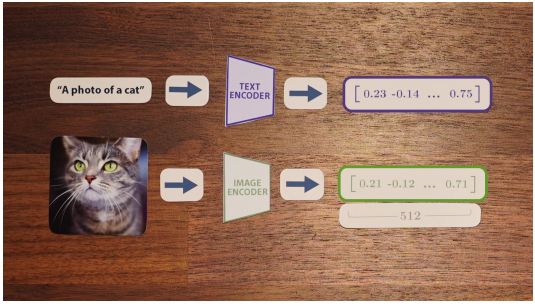
Conditioning

Classifier-free guidance

One space for words and pictures

Before we can steer with text, text and images must live somewhere in common.

CLIP: two encoders, one destination



Welch Labs / 3Blue1Brown (2025)

CLIP - Contrastive Language-Image Pre-training (Radford et al., 2021) - is two separate networks:

- a **text** encoder: caption → vector
- an **image** encoder: picture → vector

Both output vectors of the same length (512 in the original), into the same space.

The goal: an image and *its own caption* should land in nearly the same place. Trained on 400 million image-caption pairs scraped from the web.

Contrastive learning, in one idea

We have no labels saying "this vector should be at coordinates $(0.3, -1.1, \dots)$". All we have is which caption belongs to which image. That is enough.

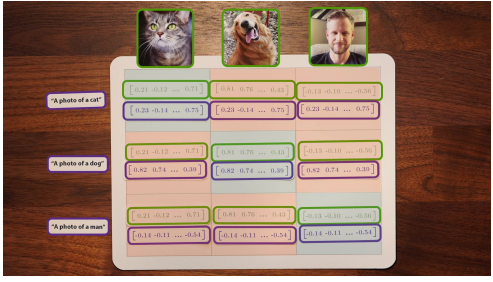
Pull together what belongs together. **Push apart** everything else.

Nobody says where anything goes. The geometry is a *consequence* of thousands of these pulls and pushes settling against each other.

Where the negatives come from is the clever part. Take a batch of N pairs. Each image has one correct caption and $N-1$ wrong ones - already in the batch, free.

Bigger batches mean harder negatives, which is one reason CLIP needed to be trained at scale.

The training objective is a matrix



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Images as columns, captions as rows. Every cell is the similarity of one image with one caption.

- the **diagonal** holds the true pairs → maximize
- everything **off-diagonal** is a mismatch → minimize

That is the whole loss: a classification problem where the "classes" are "which of these N captions is mine?", run in both directions at once.

The **C** in CLIP is this contrast.

The similarity is one you already know

CLIP scores a pair with **cosine similarity** - the cosine of the angle between the two vectors:

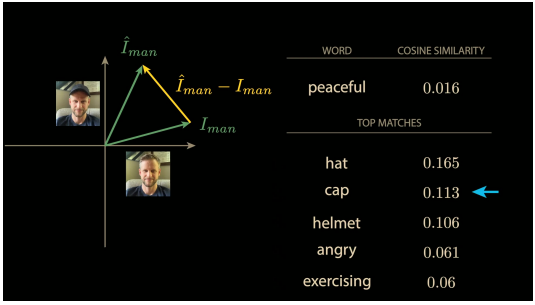
$$\mathrm{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} = \cos \vartheta$$

This is the dot product from LLM-3, normalized. There we used it to ask how much one token should attend to another; here we ask how much a caption matches a picture. Same operation, same meaning: *alignment*.

Dividing out the magnitudes means only **direction** carries meaning. Two vectors pointing the same way are "the same idea", regardless of length.

You met cosine once before as a distance option in clustering (L13). This is where it starts earning its keep.

The space learned arithmetic nobody asked for



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Encode a photo of a person **wearing a hat**. Encode the same person **without**. Subtract the two image vectors.

Now search the *text* side for the caption closest to that difference.

The top match is "**hat**" (0.165), then "cap", then "helmet".

Nothing in the training objective asked for this. Pull matches together, push mismatches apart, and a space falls out where **directions correspond to concepts** - and where you can cross from images to words and back.

Two consequences

Classification for free. Encode an image. Encode "a photo of a {cat, dog, truck, ...}". Pick the caption with the highest similarity.

No training on those classes - **zero-shot**. CLIP was competitive with supervised models on datasets it had never been fitted to.

But it only goes one way. Both encoders map *into* the space. Neither comes back out.

CLIP can tell you an image matches "a lone tree in the desert". It cannot draw one.

We have a model that turns pictures into meaning, and a model that turns noise into pictures.

Bolt them together.

Conditioning

Getting the prompt inside the network. The easy half.

Give the denoiser one more input

The network so far is $\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$. Add the prompt as a third argument:

$$\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t,\ t,\ \mathbf{c})$$

Train exactly as before, on image-caption pairs, with the same $\mathcal{L}_{\text{simple}}$ from L28.

The bet: knowing the image is *of a tree* should help the model predict which noise was added to it. Better denoising, and a model that has learned to use text.

We have done this before. Conditioning on t in L27 was the same move: an extra input that tells the network which situation it is in. Text is just a much richer t.

What exactly goes in?

The obvious answer - "the CLIP text vector" - is **not** what production models do.

A single pooled 512-d vector squeezes the whole sentence into one point. "A dog chasing a cat" and "a cat chasing a dog" become nearly identical.

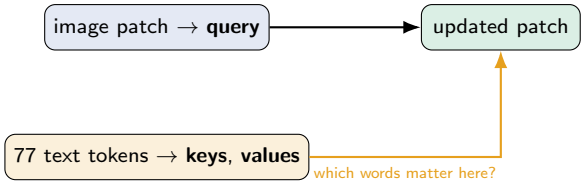
Stable Diffusion takes the layer *before* the pooling: a **sequence** of per-token vectors, $\mathbf{77} \times \mathbf{768}$. One vector per token, not one per sentence.

Why 768 and not the 512 from two slides ago: Stable Diffusion uses a larger CLIP variant (ViT-L/14, width 768) than the one in the original zero-shot demos (ViT-B/32, width 512). The 77 is CLIP's fixed maximum token count - prompts are padded or truncated to it.

This is not a detail. The mechanism on the next slide **needs** a sequence to work at all - there is nothing to look across if the prompt is a single point.

Cross-attention: how the pixels read the prompt

In LLM-3 attention let each token look at every other token, by matching a **query** against **keys** and taking a weighted sum of **values**. Same machinery here - but the queries and the keys come from different places.



Every patch of the image asks: *which words in this prompt are about me?* A patch that is becoming sky attends to "sky"; a patch becoming bark attends to "tree".

Self-attention (LLM-3): queries, keys and values all from one sequence.

Cross-attention: queries from the image, keys and values from the text. That is the only difference.

These layers sit inside the UNet blocks from L28. Simpler alternatives exist - concatenating the text vector, or adding it like the time embedding - and many models use several routes at once.

Doing it properly: DALL·E 2

OpenAI trained a diffusion model to **invert the CLIP image encoder**: given a CLIP embedding, produce an image that would encode to it.

CLIP maps pictures to meaning; the diffusion model maps meaning back to pictures. They called it **unCLIP** (Ramesh et al., 2022); the product was **DALL·E 2**.

Prompt adherence jumped far beyond anything before it. This is the moment text-to-image stopped being a demo.

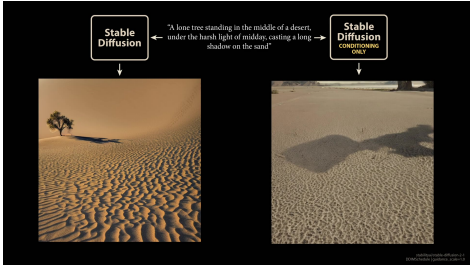
Stable Diffusion (Rombach et al., 2022) arrived around the same time from Heidelberg, worked differently in an important way - the subject of L31 - and was **open source**, which is why it is the model everyone experiments with.

Is that enough?

Predict first: we have a well-trained conditional model. Prompt: *"a lone tree standing in the middle of a desert, under the harsh light of midday, casting a long shadow on the sand."* What comes out?

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A desert. Harsh midday light. A long shadow on the sand.

No tree.

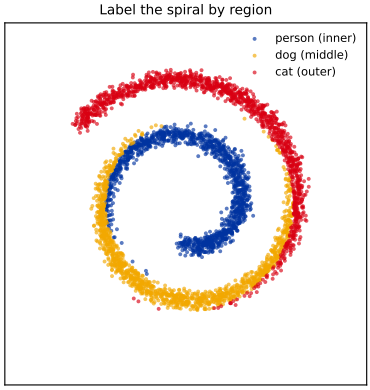
The model used the prompt - the scene is right. But it is behaving like a model that was asked for a *plausible image, given* the text, and plausible desert photographs often have no tree in them.

Conditioning tells the model what the prompt *is*. It does not make the model **care**.

Classifier-free guidance

One idea, three lines of code, and the reason your prompts work.

Back to the spiral, with labels



Split the spiral into three regions and call them classes - our stand-in for a prompt. Train exactly as in L28, with the class as an extra input.

The model now has two jobs at once: **land on the spiral** (all classes) and **land on the right part of it** (this class).

The first job is easy and dominates the loss - most of the error is in getting near the data at all. The second is a small correction on top, and it gets drowned out. That is the missing tree, in miniature.

The trick: train one model to be two models

During training, **throw the label away** on a fraction of examples (around 10-20%) and replace it with a "no class" token.

One set of weights, two behaviours:

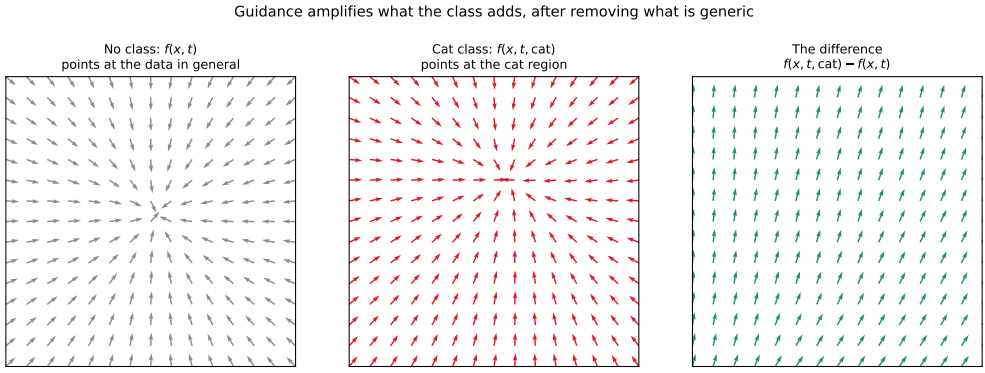
- $\varepsilon_\theta(\mathbf{x}_t, t, \mathbf{c})$ - "draw a cat"
- $\varepsilon_\theta(\mathbf{x}_t, t)$ - "draw anything"

You could train two separate networks. Dropping the label is free and gives the same thing, which is why it is universal.

Ho & Salimans (2022) - "classifier-free" because earlier work steered diffusion with a separately trained classifier, and this needs none.

Now we have both a general direction and a class-specific one, and we can compare them. **The difference between them is the part of the prediction that is about the prompt.**

Isolate the prompt's contribution, then exaggerate it



Left: without a class, the field points at the spiral in general. Middle: told "cat", it leans outward. Right: subtract them and what remains is **only** the cat-ness - the generic "get to the data" part has cancelled out.

The guidance formula

Scale that difference by α and add it back:

$$\tilde{\varepsilon} \;=\; \varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \;+\; \alpha \Big(\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{c}) - \varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \Big)$$

Use $\tilde{\varepsilon}$ in the sampler in place of the model's raw output.

Read off the special cases:

- $\alpha = 0$: ignore the prompt entirely
- $\alpha = \mathbf{1}$: exactly the plain conditional model - **no guidance**
- $\alpha > 1$: push further in the prompt's direction than the model itself would

Watch the convention. Some papers put the *conditional* prediction as the base term, which shifts what "no guidance" means. In the form above - the one `diffusers` uses - $\alpha = 1$ is neutral, and its default `guidance_scale` is 7.5.

Always check which convention a codebase means before comparing numbers.

By hand: one guidance step

One pixel. The unconditional model predicts $\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t) = 0.90$; told the prompt, it predicts $\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{c}) = 1.15$.

The prompt's own contribution is the difference:

$$1.15 - 0.90 = 0.25$$

Guidance scales *that*, and adds it back:

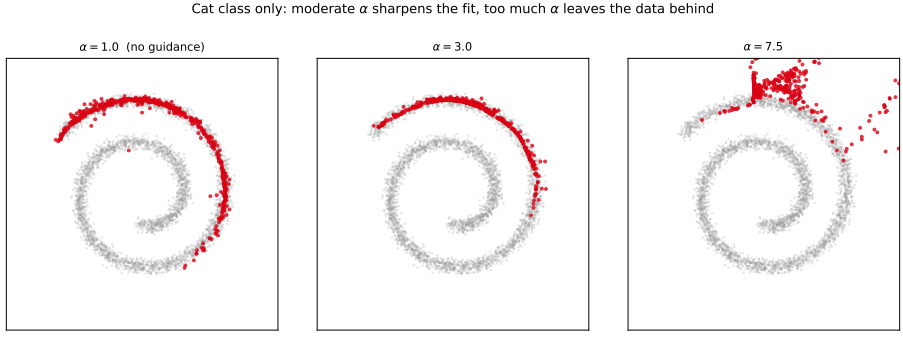
α	$\tilde{\varepsilon} = 0.90 + \alpha(0.25)$	
0	0.900	prompt ignored
1	1.150	**= the conditional model**
3	1.650	
7.5	2.775	the usual working range

Check the $\alpha = 1$ row against the formula on the last slide: the base term and the subtracted term cancel, leaving exactly $\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{c})$. **That is why $\alpha = 1$ means no guidance**, and not $\alpha = 0$.

Note how far $\alpha = 7.5$ pushes: 2.775 against a raw prediction of 1.15. We are asking the sampler to remove more than twice the noise the model actually thinks is there. It works - but you can see why it eventually stops working.

On the spiral: the cost curve

Sampling the outer class only, at three guidance strengths:

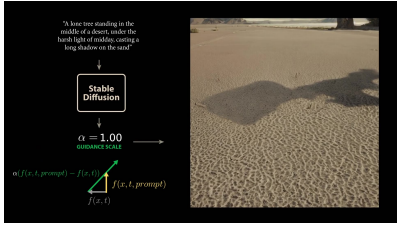


Three classes split by radius is an **easy** problem, so conditioning alone already lands on the right arm at $\alpha = 1$. The toy does not reproduce the missing tree - the Stable Diffusion image does that.

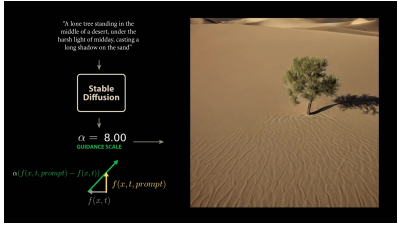
What it *does* show is overdoing it. At $\alpha = 7.5$ the samples **leave the spiral altogether** - mean radius $0.91 \rightarrow 1.10$, past the edge of the data.

And on a real model

Same prompt, same weights, same sampler. **Only α changes.**



$\alpha = 1$ - no guidance. Desert, shadow, **no tree**.



$\alpha = 8$ - the tree is there.

We changed one number. The prompt already pointed at a tree; guidance just amplified that direction.

Practically every image model you have used runs with guidance on. It is the difference between "vaguely related to your prompt" and "your prompt".

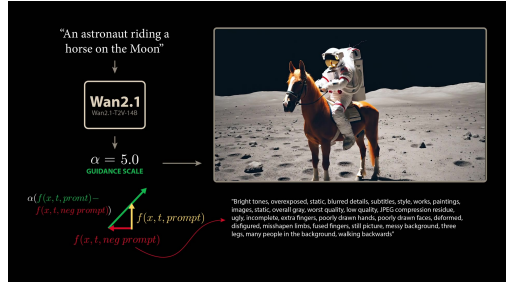
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Negative prompts: subtract what you do not want

Nothing forces the base term to be the *unconditional* prediction. Replace it with the prediction for a prompt describing what you want to **avoid**:

$$\tilde{\varepsilon} = \varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{c}_{\text{neg}}) + \alpha\big(\varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{c}) - \varepsilon_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{c}_{\text{neg}})\big)$$

Now you are steering *away* from a named region as well as toward one.



Welch Labs / 3Blue1Brown (2025)

Wan 2.1's standard negative prompt lists things like extra fingers, deformed limbs, and people walking backwards - and is written in Chinese.

The bill for all this

Guidance spends diversity to buy obedience.

High α concentrates samples in the region the prompt describes most typically. Ask for "a cat" at $\alpha = 15$ and you get the same few cats.

And it doubles the cost. Each step needs *two* forward passes - conditional and unconditional - so a guided 25-step sample is 50 network calls.

Typical values sit around 5-8. Beyond that images get oversaturated and start to look the same.

The spiral showed the far end of this: past a certain α the samples stopped resembling the data at all. On real models the same overshoot shows up as blown-out colour and hard edges rather than as points leaving a curve.

Recap

1. CLIP trains a text and an image encoder into one space, contrastively - pull true pairs together, push the rest apart.
2. The resulting geometry supports arithmetic on concepts, and zero-shot classification. But it only maps *into* the space.
3. Conditioning feeds the prompt to the denoiser - as a **sequence** of token vectors, read by **cross-attention**.
4. Conditioning alone is too weak: correct scene, missing subject.
5. Drop the label sometimes in training, and one model gives both a conditional and an unconditional prediction.
6. Amplify the difference: **classifier-free guidance**. This is why prompts work.

Next: we can now generate what we ask for.

But all of this runs in full pixel space, with two network passes per step. At 512×512 that is far beyond a laptop - and yet Stable Diffusion runs on one.

L31: how, plus video, and what all this actually costs.